MATH 423: Lecture 5

Portfolio Management

Jiamin Jian

University of Michigan

Winter 2026

ONLY FOR MATH 423 WINTER 2026
NO PERMISSION FOR UPLOADING
HAS BEEN GIVEN

Contents

(1) Method of Lagrange multipliers
(2) Several securities
(3) The Markowitz bullet / minimum variance line
(4) Efficient frontier
(5) Market portfolio
(6) Capital market line

ONLY FOR MATH 423 WINTER 2026
NO PERMISSION FOR UPLOADING
HAS BEEN GIVEN

Lagrange Multiplier (Single Constraint)

Let $f, g : \mathbb{R}^2 \to \mathbb{R}$ and suppose they are continuously differentiable. Suppose we want to solve

$$\max_{x_1, x_2} f(x_1, x_2) \text{ subject to } g(x_1, x_2) = 0 \tag{1}$$

Introduce Lagrangian $\mathcal{L}$,

$$\mathcal{L}(x_1, x_2, \lambda) := f(x_1, x_2) + \lambda g(x_1, x_2)$$

If (x_1^*, x_2^*) is a solution to (1) with $\nabla g(x_1^*, x_2^*) \neq 0$, then there exists λ^* such that $\nabla \mathcal{L}(x_1^*, x_2^*, \lambda^*) = 0$.

Lagrange multipliers

Let $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}$ and suppose they are continuously differentiable. Consider the problem

$$\min_{x \in \mathbb{R}^n} f(x), \text{ subject to } g(x) = 0. \tag{2}$$

Introduce the Lagrangian,

$$\mathcal{L}(x, \lambda) = f(x) + \lambda g(x)$$

If $x^* \in \mathbb{R}^n$ is a solution to (2) with $\nabla g(x^*) \neq 0$, then there exists λ^* such that $\nabla \mathcal{L}(x^*, \lambda^*) = 0$.

Several Securities

Now suppose that the market has n different stocks, then a portfolio can be described by weights (in these stocks)

$$w_i = \frac{x_i S_i(0)}{V(0)}, \qquad i = 1, \ldots, n,$$

where x_i denotes the position in stock i and $V(0) = \sum_{i=1}^{n} x_i S_i(0)$.

- Obviously, $\sum_{i=1}^{n} w_i = w_1 + \cdots + w_n = 1$. (Two securities: $w_1 + w_2 = 1$)
- Let K_i be the return of stock i, then we have

$$K_V = w_1 K_1 + \cdots + w_n K_n = \sum_{i=1}^{n} w_i K_i.$$

(Two securities: $K_V = w_1 K_1 + w_2 K_2$)

ONLY FOR MATH 423 WINTER 2026
NO PERMISSION FOR UPLOADING
HAS BEEN GIVEN

Def 5.1

(a) Define weight row vector $w := (w_1, \ldots, w_n)_{1\times n}$ and $u := (1, \ldots, 1)_{1\times n}$, thus $uw^T = (1, \ldots, 1)(w_1, \ldots, w_n)^T = 1$;

(b) Define the mean vector $m := (\mu_1, \ldots, \mu_n)_{1\times n}$ where $\mu_i := \mathbb{E}(K_i)$;

(c) Define the $n \times n$ covariance matrix $C = (c_{ij})_{n\times n}$, where $c_{ij} :=$ Cov(K_i, K_j).

Rmk 5.1

(a) C is symmetric (the transpose $C^T = C$) and positive definite ($XCX^T \geq 0$, for any real vector $X = (x_1, \cdots, x_n)_{1\times n} \in \mathbb{R}^n$);

(b) The diagonal elements $c_{ii} =$ Cov$(K_i, K_i) =$ Var$(K_i) = \sigma_i^2$;

(c) In what follows, we shall assume C has the inverse C^{-1}.

Prop 5.1

The expected return $\mu_V = \mathbb{E}(K_V)$ and variance $\sigma_V^2 = \text{Var}(K_V)$ of a portfolio with weights w are given by

$$\mu_V = mw^T = (\mu_1, \cdots, \mu_n)(w_1, \cdots, w_n)^T,$$
$$\sigma_V^2 = wCw^T.$$

Proof: See Capinski Prop 3.23.

Eg 5.1

Given $(w_1, w_2, w_3) = (50\%, -25\%, 75\%)$; $(\mu_1, \mu_2, \mu_3) = (10\%, 7\%, 12\%)$; $\sigma_1 = 1,\ \sigma_2 = 0.5,\ \sigma_3 = 2$; and $\rho_{12} = -0.5,\ \rho_{13} = 0.5,\ \rho_{23} = 0$, find the expected return μ_V and the risk σ_V of the portfolio.

We can get $\mu_V = \mathbb{E}(K_V) = mw^T = 0.1225$ and

$$C = \begin{bmatrix} 1 & -0.25 & 1 \\ -0.25 & 0.25 & 0 \\ 1 & 0 & 4 \end{bmatrix} \Rightarrow \sigma_V^2 = wCw^T = \frac{71}{32} \Rightarrow \sigma_V = 1.4895.$$

More on the σ, μ plane (sometimes called the risk-expected return graph):

(a) Given a weight vector w of a portfolio, we can calculate the pair μ_V and σ_V by the last proposition. (Actually $\mu_V(w)$ and $\sigma_V(w)$)

(b) We can draw the pair (μ_V, σ_V) on the σ, μ plane. When w moves to $\tilde{w}$, the point $(\mu_V(w), \sigma_V(w))$ moves to a new location $(\mu_V(\tilde{w}), \sigma_V(\tilde{w}))$ in the σ, μ plane.

(c) The collection of all such points is called the "Area of the Portfolio" (or feasible region, or attainable sets), we shadow out this region and its boundary is called "Markowitz bullet". (Why the area of portfolio can not cover the whole σ, μ plane? $ww^T = 1$.)

(d) However in the two-stock case, the area is just a simple curve. i.e. all portfolios lie on the curve, not what inside.

(e) We see that the minimal risk is attained at the vertex (point) of the shadowed area, which is exactly located on the "Markowitz bullet". There must be a portfolio corresponding to the vertex. We call it the "minimal variance portfolio", and denote its weight by w^*.

ONLY FOR MATH 425 WINTER 2006
NO PERMISSION FOR UPLOADING
HAS BEEN GIVEN

Definition 5.2

The portfolio with the smallest variance in the attainable set is called the minimum variance portfolio (MVP).

Proposition 5.2 (Weights of the MVP)

The weights of the minimum variance portfolio are given by

$$w_{\text{MVP}} = \frac{uC^{-1}}{uC^{-1}u^T}.$$

Remark 5.2

Recall that, in the case of $n = 2$, $w_1^{\min} = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$ and $w_2^{\min} = 1 - w_1^{\min}$.

Check:

$$C = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix}, \quad C^{-1} = \frac{1}{\det(C)} \begin{bmatrix} \sigma_2^2 & -\sigma_{12} \\ -\sigma_{12} & \sigma_1^2 \end{bmatrix}$$

and

$$\det(C) * uC^{-1} = (\sigma_2^2 - \sigma_{12}, \sigma_1^2 - \sigma_{12}), \quad \det(C) * uC^{-1}u^\top = \sigma_2^2 + \sigma_1^2 - 2\sigma_{12}.$$

Proof: We want to solve the problem

$$\min_{w} wCw^\top \text{ subject to the constraint } \sum_{i=1}^{n} w_i = uw^\top = 1.$$

Since we have a minimization problem with constraints, we will use the Lagrange multipliers method. To this end, define

$$\mathcal{L}(w, \lambda) := wCw^\top - \lambda(uw^\top - 1).$$

Then, for $\nabla := (\frac{\partial}{\partial w_1}, \ldots, \frac{\partial}{\partial w_n})$, we get ☼

$$\nabla\mathcal{L} = 0 \Leftrightarrow 2wC - \lambda u = 0 \Leftrightarrow w = \frac{\lambda}{2} uC^{-1}. \tag{3}$$

Substitute (3) in the constraint in order to determine the λ. This provides

$$1 = uw^\top = \frac{\lambda}{2} u\big(uC^{-1}\big)^\top = \frac{\lambda}{2} uC^{-1}u^\top \Leftrightarrow \lambda = \frac{2}{uC^{-1}u^\top}. \tag{4}$$

Finally, use (3) & (4) in order to derive the weights

$$w_{\text{MVP}} = \frac{uC^{-1}}{uC^{-1}u^\top}.$$

This is a necessary condition. Since the function $wCw^\top$ is quadratic and non-negative, this constraint problem must have a minimum value.

ONLY FOR MATH 423 WINTER 2026
NO PERMISSION FOR UPLOADING
HAS BEEN GIVEN

The Markowitz bullet / Minimum Variance Line

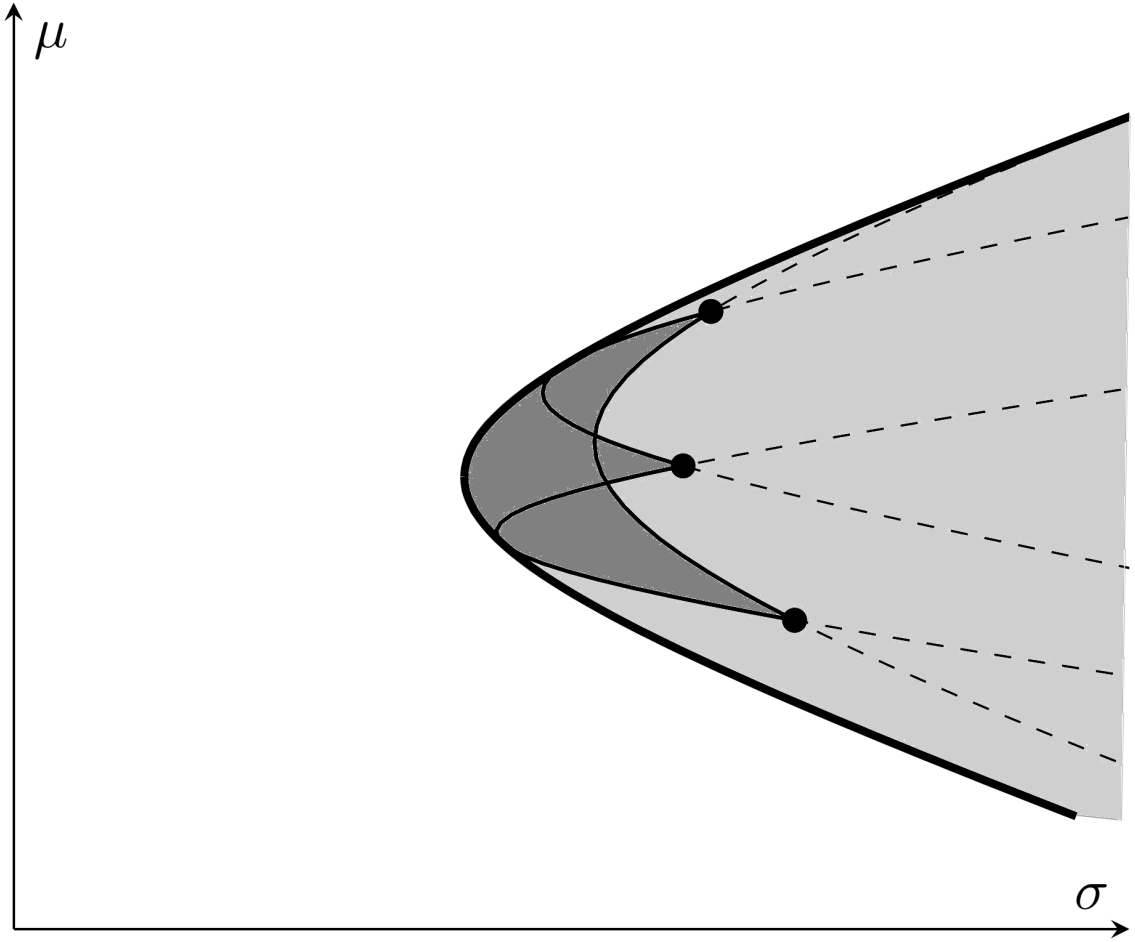


Figure: The attainable set and the Markowitz bullet (or minimum variance line)

Remark 5.3

The bold lines of the last pictures are called the Markowitz bullet; it was named after Harry Markowitz in conjunction with its shape ("bullet").

An important feature of the Markowitz bullet is that each point on it has the smallest risk among all portfolios with the same expected return as the point.

This is why the Markowitz bullet is also called the minimum variance line.

Definition 5.3

The set (parameterized on the positive values of the expected return) of the portfolios with minimum risk for a given level of expected return is called the Minimum Variance Line (MVL).

Proposition 5.3 (Weights of portfolios on the MVL)

Fix a level of expected return μ_V. The portfolio with the smallest risk among all portfolios with expected return μ_V has weights

$$w_{\mu_V}^{\min} = \frac{\begin{vmatrix} 1 & c \\ \mu_V & d \end{vmatrix} uC^{-1} + \begin{vmatrix} a & 1 \\ b & \mu_V \end{vmatrix} mC^{-1}}{\begin{vmatrix} a & c \\ b & d \end{vmatrix}} = \frac{(d - c\mu_V)uC^{-1} + (a\mu_V - b)mC^{-1}}{ad - bc}$$

(Recall $\begin{vmatrix} a & c \\ b & d \end{vmatrix} = ad - bc$) Here $a = uC^{-1}u^T$, $b = mC^{-1}u^T$, $c = uC^{-1}m^T$, $d = mC^{-1}m^T$ are all real numbers.

Note: a, b, c, d are all constants independent of the weight vector w.

Remark 5.4

$w_{\mu_V}^{\min}$ is linear in μ_V:

$$w_{\mu_V}^{\min} = A\mu_V + B$$

with row vectors $A = \dfrac{(am - cu)C^{-1}}{ad - bc}$ and $B = \dfrac{(du - bm)C^{-1}}{ad - bc}$.

Proof: We want to solve the problem

$$\min_w wCw^\top \text{ subject to the constraints } uw^\top = 1 \;\&\; mw^\top = \mu_V.$$

Similarly to the Proposition 5.2, we will use the Lagrange multipliers method. Define

$$\mathcal{L}(w, \lambda) := wCw^\top - \lambda_1(uw^\top - 1) - \lambda_2(mw^\top - \mu_V).$$

Then ☼

$$\nabla\mathcal{L} = 0 \Leftrightarrow 2wC - \lambda_1 u - \lambda_2 m = 0 \Leftrightarrow w = \frac{1}{2}(\lambda_1 u + \lambda_2 m)C^{-1} \tag{5}$$

which in conjunction with the constraints leads to the system with unknowns the elements of λ:

$$\begin{cases} \lambda_1 uC^{-1}u^\top + \lambda_2 uC^{-1}m^\top = 2, \\ \lambda_1 mC^{-1}u^\top + \lambda_2 mC^{-1}m^\top = 2\mu_V, \end{cases}$$

whose solution can be immediate determined, *e.g.*, by Cramer's rule. Then, we derive immediately $w_{\mu_V}^{\min}$ by substituting in (5).

ONLY FOR MATH 423 WINTER 2026
NO PERMISSION FOR UPLOADING
HAS BEEN GIVEN

Recall that $w_{\mu_V}^{\min} = A\mu_V + B$ for some constant vectors A and B.

Assume that you have determined two (arbitrary, but distinct) portfolios on the MVL. Then, you can simply "properly split" your wealth and invest on these two portfolios in order to obtain a portfolio with specific data (σ, μ) on MVL.

Theorem 5.1 (Two-Fund Theorem)

Choose arbitrarily two distinct portfolios on the MVL, say V_1^{MVL} and V_2^{MVL}, with associated weights w_1 and w_2. Then, a portfolio V lies on the MVL if and only if its weight vector w can be written as an affine combination of w_1 and w_2, *i.e.*

$$w = \alpha w_1 + (1-\alpha) w_2, \quad \text{for some } \alpha \in \mathbb{R}.$$

Proof: Use $\mu_1 \neq \mu_2$ and Remark 5.4.

Minimum Variance Line & Efficient Frontier

Question: Based on what criterion should we make our choice among all the attainable portfolios?

Answer: Let V_i, $i = 1, 2$ be two risky assets, *e.g.*, stock price, portfolio *etc.*, with associated data (σ_i, μ_i), $i = 1, 2$. Then,

1. If $\mu_1 = \mu_2$, then we "naturally" prefer the asset with the lower risk.
2. If $\sigma_1 = \sigma_2$, then we "naturally" prefer the asset with the higher expected return.
3. If $\sigma_1 \leq \sigma_2$ and $\mu_1 \geq \mu_2$, then we "naturally" prefer the asset V_1.
4. If $\sigma_1 \leq \sigma_2$ and $\mu_1 \leq \mu_2$, *i.e.*, high risk-high returns vs. low risk-low returns, then depends on the profile of the investor.

MATH 423 WIN FEB 2026
NOT FOR UPLOADING
SEE LOVEN

Definition 5.4

(1) We say that stock (security, or portfolio) 1 dominates (or is more preferable than) stock (security, or portfolio) 2 if stock 1 has higher expected return and less risk than stock 2: $\mu_1 \geq \mu_2$ and $\sigma_1 \leq \sigma_2$.

On the σ, μ plane, (σ_1, μ_1) is "above and to the left of" (σ_2, μ_2).

(2) A portfolio is called efficient if it is not dominated by any other portfolio in the attainable set. The set of all efficient portfolios is called the efficient frontier.

Efficient Frontier

Question: Where is the efficient frontier on the σ, μ plane?

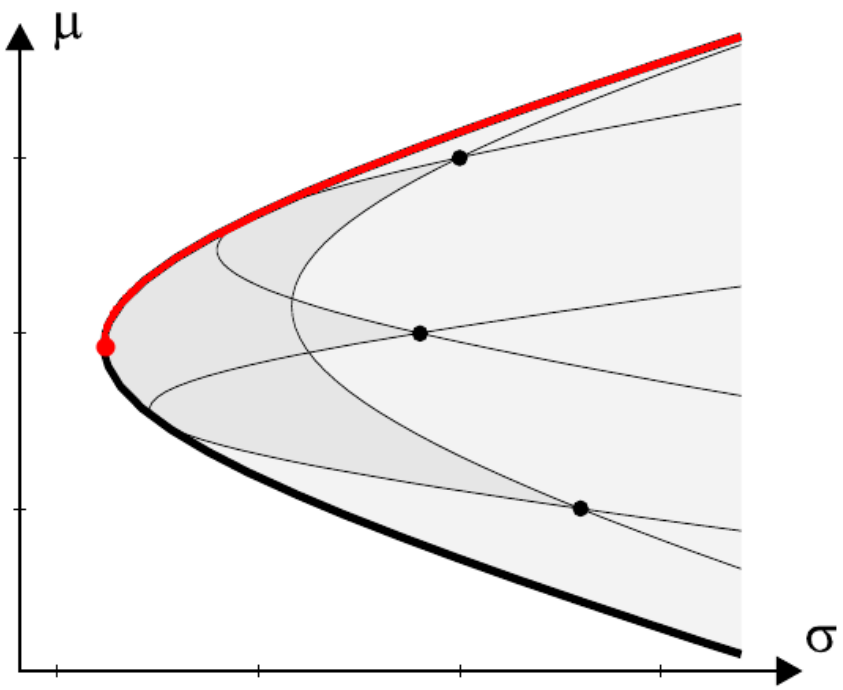


Figure: The Efficient Frontier in the example we considered before.

Note: from the graph, we see that the effective frontier consists of the upper part of the M.B. including the vertex.

Each efficient portfolio has the highest expected return among all attainable portfolios with the same risk, and has the lowest risk among all attainable portfolios with the same expected return.

For our convenience, we will denote the weights of the portfolios lying on the efficient frontier by w_{EF}, instead of $w_{\mu_V}^{\min}$, for $\mu_V \geq \mu_{\text{MVP}}$.

The next proposition provides a property of the efficient frontier which will be useful in the Capital Asset Pricing Model (CAPM), in the next lecture.

Corollary 5.1

The efficient frontier is the set of points whose weights w_{EF} are given by

$$w_{\text{EF}} = A\mu + B, \quad \text{for } \mu \geq \mu_{\text{MVP}},$$

where A, B as introduced in Remark 5.4 and μ_{MVP} is the expected return of the Minimum Variance Portfolio (MVP).

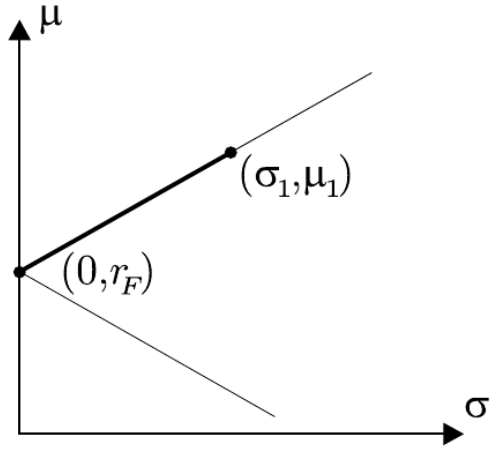
Assumption 5.1

From now on we extend the market of the n risky assets by introducing a risk-free asset with return R.

Moreover, we will assume that $R < \mu_{\text{MVP}}$. (Why?)

Market Portfolio

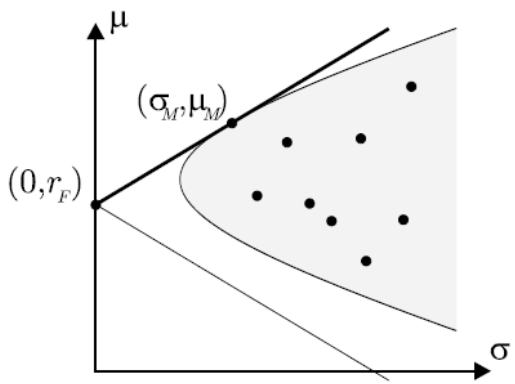
Recall the graph of the set of portfolios that we can construct based on two assets; one risk free and one risky.



Question: Among all attainable portfolios, which one should we choose, say V, in order to construct a portfolio consisting of the risk-free asset and the risky asset V?

Answer: Let's observe initially the following:

1. It is immediate that the portfolio V should lie on the efficient frontier. (Why?)
2. We should choose the portfolio V so that the half-line starting at $(0, R)$ and passing through V is tangent to the efficient frontier. (Why?)



CMU 70-361/761 WINTER 2024
NOTE PERMISSION FOR UPLOADING
NOT HAS BEEN GIVEN

In other words, we can uniquely identify a portfolio M, which will be given the special name Market Portfolio.

Proposition 5.4

Under Assumption 5.1, *i.e.*, $R < \mu_{\text{MVP}}$, and if C is invertible, then the market portfolio M exists and its weights are given by

$$w_M = \frac{(m - Ru)C^{-1}}{(m - Ru)C^{-1}u^\top}.$$

Proof: We need to solve a constrained maximization problem:

$$\max_w \frac{mw^\top - R}{\sqrt{wCw^\top}} \text{ subject to the constraint } uw^\top = 1.$$

To this end we define the Lagrangian

$$\mathcal{L}(w, \lambda) := \frac{mw^\top - R}{\sqrt{wCw^\top}} - \lambda(uw^\top - 1).$$

By following the proper steps, we derive

$$\nabla\mathcal{L} = 0 \Leftrightarrow \frac{m[wCw^\top]^{\frac{1}{2}} - (mw^\top - R)[wCw^\top]^{-\frac{1}{2}}wC}{wCw^\top} - \lambda u = 0$$

$$\Leftrightarrow m - \lambda[wCw^\top]^{\frac{1}{2}}u = \frac{mw^\top - R}{wCw^\top}wC. \tag{6}$$

ONLY FOR MAT 1423 WITH PERMISSION FOR USE HAS BEEN GIVEN

By multiplying with $w^\top$ on the right and using the constraint, we get

$$\lambda = \frac{R}{[wCw^\top]^{\frac{1}{2}}}.$$

By (6) we derive, now that we know the value of λ,

$$\frac{mw^\top - R}{wCw^\top} w = (m - Ru)C^{-1}. \tag{7}$$

The LHS is not easy to be solved. For notational simplicity, let us define $\gamma := \dfrac{mw^\top - R}{wCw^\top}$; observe that γ depends on w.

However, multiplying (7) on the right of by $u^\top$ and using the constraint, we get

$$\gamma = (m - Ru)C^{-1}u^\top; \tag{8}$$

observe that the RHS is independent of w now.

At this point, we have to make sure that $\gamma \neq 0$. Recall that

$$\mu_{\text{MVP}} = w_{\text{MVP}} m^\top = \frac{uC^{-1}m^\top}{uC^{-1}u^\top} = \frac{mC^{-1}u^\top}{uC^{-1}u^\top} \overset{\text{Assumption 5.1}}{>} R$$
$$\Leftrightarrow mC^{-1}u^\top - RuC^{-1}u^\top > 0 \Leftrightarrow \gamma > 0.$$

Combine (7) and (8) to get the required form.

ONLY FOR MATH 123 WINTER 2026
NO PERMISSION FOR UPLOADING
HAS BEEN GIVEN

Definition 5.5

The half-line that starts at the risk-free asset and runs through the market portfolio M will be said the Capital Market Line (CML) and it satisfies the equation

$$\mu = R + \frac{\mu_M - R}{\sigma_M}\sigma.$$

The quantity $\frac{\mu_M - R}{\sigma_M}$ will be said the risk premium.

Remark 5.5

1. The risk premium can be seen as the additional return above the risk-free return R which compensates for exposure to risk.
2. If every investor uses the same m and C and shares the same principles as the one we have described, *e.g.*, the efficient portfolios are preferable, then every investor will construct their portfolio on the CML.

Example

Exercise 5.1

Assume the following data: $m = (\mu_1, \mu_2, \mu_3) = (20\%, 13\%, 17\%)$; $\sigma_1 = 0.25,\ \sigma_2 = 0.28,\ \sigma_3 = 0.20$; and $\rho_{12} = 0.30,\ \rho_{13} = 0.15,\ \rho_{23} = 0.00$.

1. Find the minimum variance portfolio (MVP). Determine μ_{MVP} and σ_{MVP}.
2. Among all feasible portfolios with expected return $\mu_V = 20\%$, find the portfolio with the minimum risk. Compute the weights and the standard deviation of this portfolio.
3. Suppose that $R = 5\%$. Compute the weights in the market portfolio. Determine also μ_M and σ_M.

$$C = \begin{bmatrix} 0.0625 & 0.021 & 0.0075 \\ 0.021 & 0.0784 & 0 \\ 0.0075 & 0 & 0.04 \end{bmatrix}, \quad C^{-1} = \begin{bmatrix} 18.0282 & -4.8290 & -3.3803 \\ -4.8290 & 14.0486 & 0.9054 \\ -3.3803 & 0.9054 & 25.6338 \end{bmatrix}$$

1. Use Prop 5.2 to get $w_{\text{MVP}} = (0.2278, 0.2349, 0.5373)$, $(\mu_{\text{MVP}}, \sigma_{\text{MVP}}) = (0.1674, 0.1523)$.

2. Use Prop 5.3 to get $w_{0.2}^{\min} = (0.7221, -0.2084, 0.4863)$, $\sigma_{0.2}^{\min} = 0.2107$.

3. Use Prop 5.4 to get $w_M = (0.3778, 0.1004, 0.5218)$, $(\mu_M, \sigma_M) = (0.1773, 0.1586)$.

ONLY FOR MA
NO PERMISSIO
HAS BE